When Does a Foundation Model Help? Process-Specific Reliability of Machine-Learned Interatomic Potentials

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Abstract

Foundation models promise to make accurate atomistic simulation accessible with little material-specific training. Their practical value, however, depends on whether they reproduce the process for which a simulation is needed. Here we use Cr-doped Sb_2Te_3 to compare specialist, zero-shot and fine-tuned interatomic potentials across thermal configurations, dopant migration and collective layer sliding. Fixed-geometry energies, model-relaxed pathways and molecular dynamics expose complementary failure modes. Target adaptation improves force accuracy and selected migration-energy comparisons, yet does not ensure stable dynamics or reliable energies under stronger distortions. Conversely, zero-shot checkpoints sustain longer trajectories while remaining poorly calibrated on some migration configurations. Relaxing a pathway can lower its apparent barrier without recovering the reference energy profile. Structural projections and error surrogates help characterize these differences, but do not establish their microscopic cause. These results motivate process-specific model selection: a useful potential must be tested both on the configurations relevant to an observable and on the configurations its own forces generate. Foundation initialization is therefore a resource to validate and adapt, rather than a guarantee of downstream reliability.

Keywords: interatomic potentials, foundation models, fine-tuning, migration, distribution shift, model validation

1 Introduction

Foundation models are changing how atomistic simulations are built. Rather than assembling a new potential for every material, researchers can start from a model trained across diverse chemical environments and use it directly or adapt it with a smaller reference dataset. Equivariant machine-learned interatomic potentials (MLIPs) have made this prospect especially attractive: they connect first-principles energies and forces to simulations at scales that are difficult to reach with density functional theory (DFT) alone [1–3]. Broadly trained potentials such as MACE-MP and CHGNet extend this approach across materials [4, 5]. The resulting choice is increasingly practical: should a researcher use a foundation checkpoint, fine-tune it, or train a specialist?

For a materials scientist, the answer must ultimately be expressed in terms of a physical process. A model may reproduce typical thermal configurations yet misrepresent the short-lived coordination changes that control dopant motion. It may remain numerically stable while exploring an inaccurate energy landscape. Conversely, a small energy error on a supplied transition path does not guarantee that the model's forces will generate that path. These distinctions matter because activated motion depends on relative energies along rare configurations, while a simulated trajectory repeatedly uses the model's own predictions to determine what it sees next. Average error over a test set and reliability in a target process are therefore different questions.

Recent work establishes both the promise and the conditional nature of transfer. Alghamdi and colleagues show that zero-shot and lightly fine-tuned MACE models can reproduce Li diffusion observables in LiF; Bheemaguli and colleagues distinguish migration-energy accuracy from transition-path geometry across battery materials; and Deng and colleagues identify systematic softening of universal potentials on several out-of-distribution tasks [6–8]. These studies make process-aware testing essential. They also suggest that the success of adaptation depends on which parts of the energy landscape the target data constrain. What remains useful to resolve is how the verdict changes when the same material is examined through different processes and different evaluation procedures.

We address this question using Cr-doped Sb_2Te_3. Its quintuple layers comprise a Te–Sb–Te–Sb–Te stack, separated by van der Waals gaps. A Cr atom moving within a gap, approaching a layer or entering its interior encounters different coordination environments. Translating one layer relative to another introduces a collective registry change. This combination supplies a compact set of physically distinct probes within one host (Fig. 1), without treating a change of material as the only form of distribution shift. Cr-doped Sb_2Te_3 is also an established platform for studying the interplay of transition-metal doping and layered topological materials [9, 10].

The comparison separates two sources of difficulty. The first is a change in the configurations being evaluated, from thermal fluctuations to migration and sliding. The second is a change in how configurations are obtained: a common reference path tests energy assignments, whereas an MLIP-relaxed band or an MD trajectory tests the feedback between predictions and subsequent configurations. This distinction makes it possible to ask what a foundation model helps with, where adaptation helps, and which apparent improvements fail to carry over to another use.

An earlier report introduced migration-based comparisons for this system and included sliding and representation analyses [11]. The present study revisits those conclusions with trajectory-grouped evaluations, additional training seeds, explicit fixed-path and self-relaxed operators, a broader checkpoint comparison, and corrected diagnostic interpretations. Its contribution is a more discriminating test of reliability, rather than a universal ranking of training strategies. We find that foundation models can supply useful dynamical robustness, but neither foundation initialization nor target fine-tuning guarantees the energy landscape required for a chosen process.

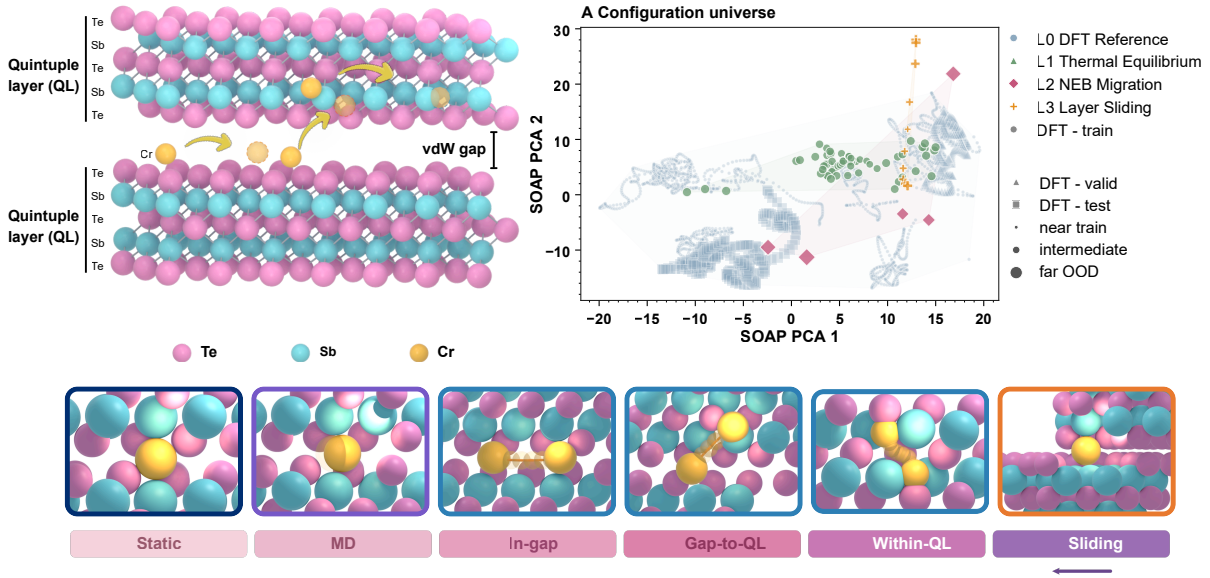


Fig. 1 A material contains several distinct tests of an interatomic potential. a, Cr-doped Sb_2Te_3, showing the Te–Sb–Te–Sb–Te quintuple layers (QLs), the van der Waals gap and representative dopant displacements. Te, Sb and Cr are pink, cyan and yellow, respectively. **b,** SOAP-based principal-component visualization of reference configurations, thermal trajectories, migration images and layer-sliding configurations. Symbols distinguish configuration sources and reference split assignments; the projection is a descriptive view of sampled configurations, not a calibrated measure of extrapolation risk. **c–h,** Local views of static configurations, molecular dynamics, in-gap migration, gap-to-QL motion, within-QL motion and sliding, in that order. The task categories organize the tests; their ordering does not define a universal or quantitatively monotonic distribution-shift scale.

2 Results

2.1 Pointwise accuracy gives different answers for energies and forces

We first compare four trained conditions: Scratch-600K, FT-600K, Scratch-MultiT and FT-MultiT. Each condition is represented by training seeds 123, 234 and 345. The 600 K families share a grouped training partition of 1,786 configurations and a test partition of 267 configurations; the MultiT families use 2,659 training and 767 test configurations. Complete source trajectories remain within a single partition. Figure 2 separates training and test predictions so that agreement on the training distribution does not obscure held-out performance.

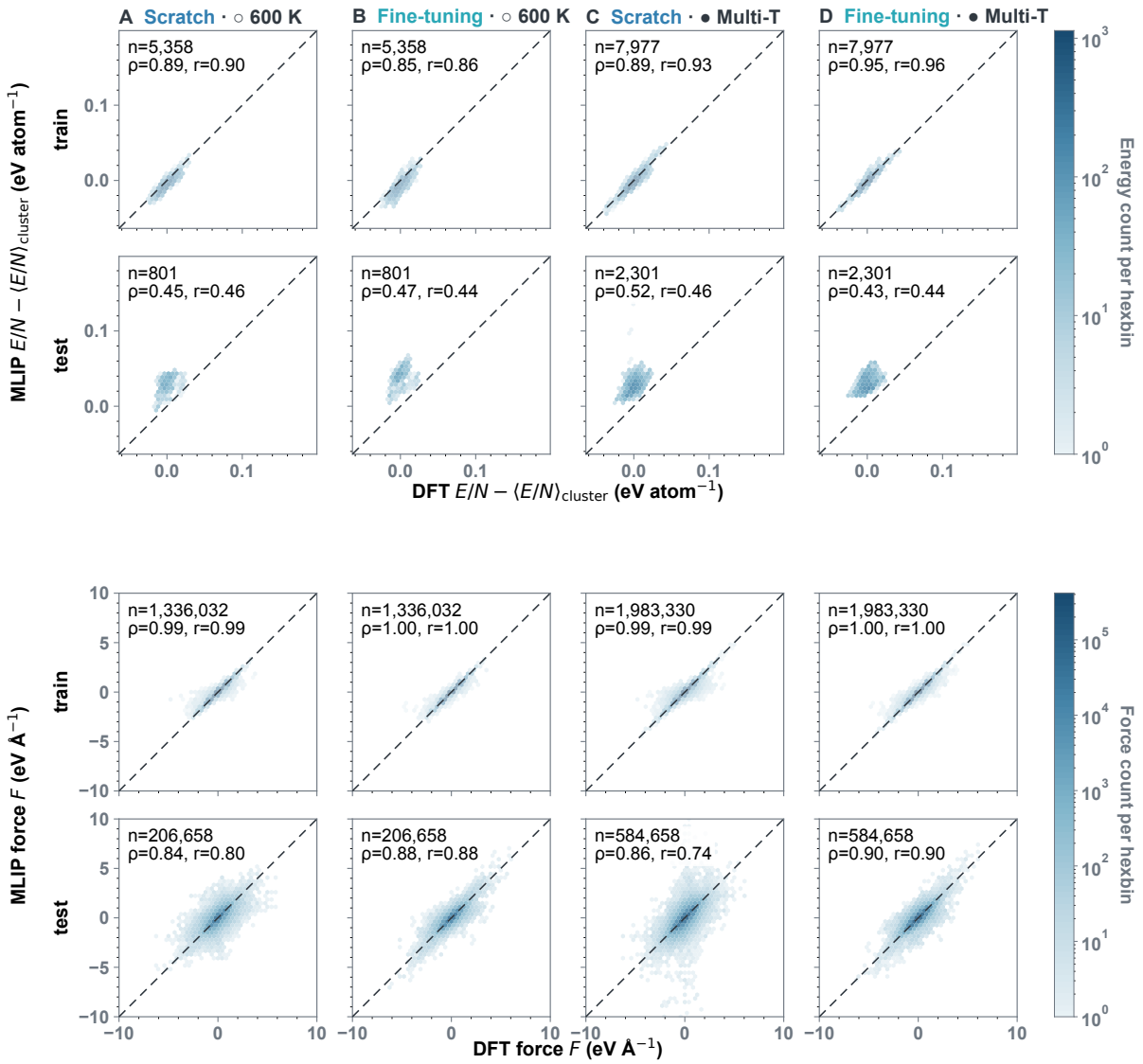


Fig. 2 Training and held-out pointwise agreement across four model conditions. a–d, Columns show Scratch-600K, FT-600K, Scratch-MultiT and FT-MultiT, respectively. Rows show training energies, test energies, training force components and test force components. Energies are per atom and are centred by subtracting the pooled DFT mean for the corresponding trajectory cluster from both DFT and MLIP values; test-only cluster means are used for visualization only. Forces are uncentred Cartesian components. Predictions from three training seeds are pooled, so the displayed n counts prediction records rather than independent structures: 5,358/801 training/test energy records for each 600 K family and 7,977/2,301 for each MultiT family. Hexbin colour denotes count on logarithmic scales, with separate normalizations for energy and force. Dashed lines indicate identity; ρ and r are Spearman and Pearson correlations. The force display is restricted to ± 10 eV Å^{-1}, leaving 106 Scratch-MultiT test components outside the plotted window; those components remain in all numerical metrics. Uncentred, seed-resolved errors are given in Supplementary Table S1.

Fine-tuning gives smaller test force errors than training from scratch in both temperature conditions. The energy errors show a different ordering. On the 600 K test partition, FT-600K has an energy RMSE of 40.68 ± 4.88 meV atom^{-1} and a force RMSE of 259.68 ± 5.83 meV Å^{-1}, compared with 30.48 ± 3.48 meV

atom^{-1} and 343.55 ± 36.27 meV Å^{-1} for Scratch-600K (Supplementary Table S1). These are means and sample standard deviations over training seeds. Adaptation helps the force observable, while the energy observable favours another route.

The energy panels display fluctuations after subtracting a common DFT reference mean within each trajectory cluster from both axes. This subtraction preserves prediction residuals, but the plotted correlation is a within-cluster diagnostic. Absolute, uncentred RMSE is reported separately. Neither energy correlation nor force RMSE specifies how accurately a model will describe a rare coordination change, and neither tests the consequences of letting predicted forces generate new configurations.

2.2 Stable trajectories and accurate dynamics are separate requirements

The MD comparison makes the distinction between local fit and model-driven evolution explicit (Fig. 3). We examine three zero-shot foundation checkpoints and three seeds for each of the four trained conditions. All runs start from the same archived 2,050-atom state at a simulation clock of 200 ps and target a total clock of 1 ns. The common continuation therefore spans at most 800 ps. Time is reconstructed from the recorded step and 1 fs integration timestep, and each trajectory is retained only to its actual endpoint; failed continuations are not treated as completed nanosecond trajectories.

All three foundation checkpoints reach the target clock. Only FT-600K seed 234 does so among the six fine-tuned models. The other two FT-600K runs end at 236 and 485 ps, and the FT-MultiT runs end between 237.2 and 286 ps. All six scratch runs reach a scientific stopping threshold between 200.4 and 204.4 ps (Supplementary Table S8). These are observed outcomes under one common starting state and thermostat protocol, not estimates of failure probabilities over independent thermal replicas. In this setting, the broader prior provides a longer usable simulation window than the tested specialist and most adapted checkpoints.

Survival alone does not establish dynamical accuracy. Temperature, pressure, in-plane mean-square displacement (MSD), coordination and velocity autocorrelation probe different aspects of the generated trajectories. The foundation trajectories show greater mobility in the displayed MSD comparison and a different local Cr environment, even where the all-atom radial distribution functions retain similar principal peaks. A global structural average can therefore conceal a local dopant discrepancy. The thermostat can also regulate kinetic temperature while other indicators reveal instability.

The truncated trajectories make a common long-time diffusion ranking inappropriate. A short nearly immobile trace is not evidence of a physically small diffusion coefficient when it terminates at a stopping threshold. We use the recorded MD observables to distinguish observed stability, local structure and mobility, and do not infer equilibrium transport coefficients or thermal conductivity from this comparison.

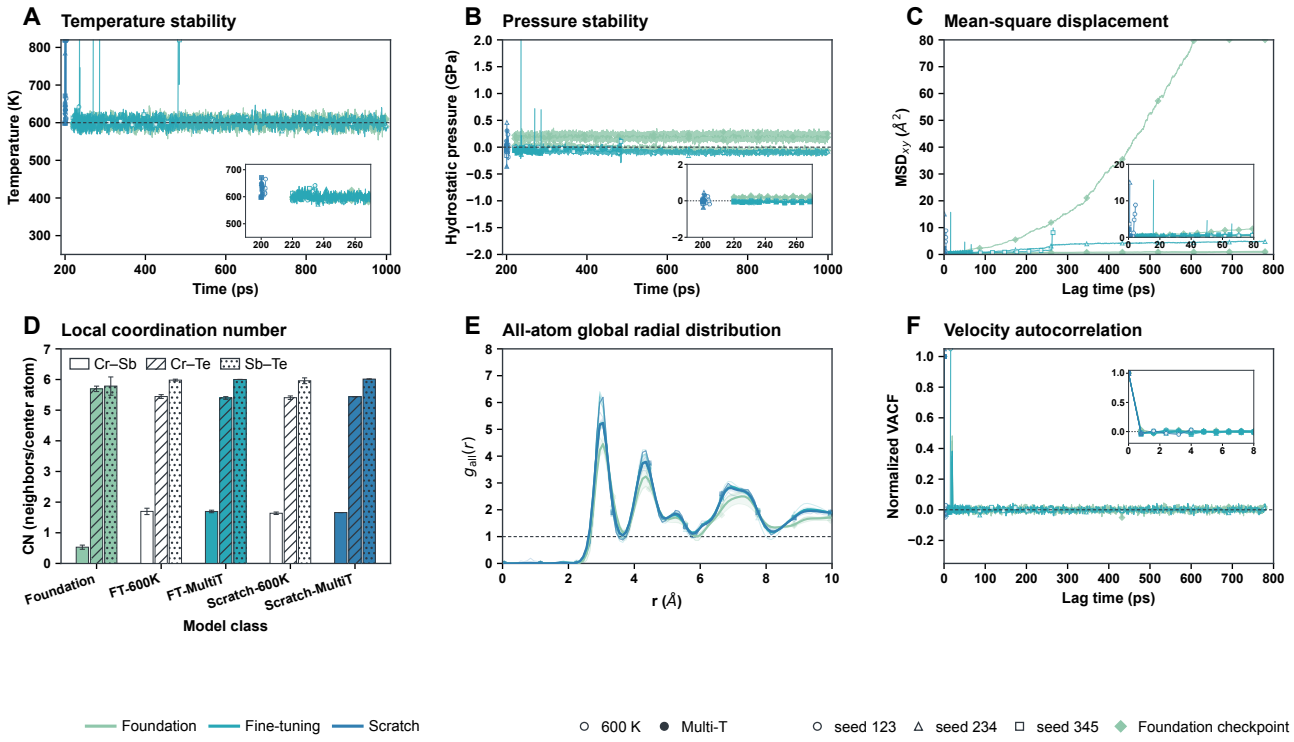


Fig. 3 Molecular dynamics tests the configurations generated by each model. The comparison contains MPA-0, MP-0b3 and OMAT-0 foundation checkpoints and seeds 123, 234 and 345 for FT-600K, FT-MultiT, Scratch-600K and Scratch-MultiT. **a,** Kinetic temperature. **b,** Hydrostatic pressure, $P = -\mathrm{tr}(\sigma)/3$, from archived stress tensors. **c,** In-plane, multiple-origin MSD with triclinic periodic unwrapping. **d,** Pair-resolved coordination numbers, using fixed Cr–Sb, Cr–Te and Sb–Te cutoffs of 3.2, 4.1 and 4.2 Å, respectively; bars and whiskers summarize the mean and sample standard deviation within each displayed condition. **e,** All-atom RDFs from ordered pair counts and triclinic minimum-image distances; thin traces retain individual models and the summaries use display-only smoothing. **f,** Normalized multiple-origin velocity autocorrelation. Insets expand early-time behaviour. Colours distinguish strategies, open/filled symbols distinguish 600 K/MultiT training, and markers identify training seeds; foundation differences are checkpoint variation. Runs start at a common 200 ps state and target 1,000 ps, using a 1 fs timestep and 200-step output interval. Curves end at their observed horizons; structural summaries use the available, unequal trajectory windows. Only three foundation checkpoints and FT-600K seed 234 reach the target (Supplementary Table S8).

2.3 A correct energy assignment does not guarantee a correct pathway

Migration probes the relative energies along configurations that connect local environments. We distinguish fixed-path scoring, in which all models see the same reference geometries, from endpoint-initialized NEB, in which each model moves the intermediate images with its own forces. These operators measure energy calibration on a supplied path and the outcome of model-driven path optimization, respectively; their peak energies cannot be interchanged or used as evidence for the same physical barrier without checking the resulting geometries and reference convergence.

Figure 4 shows this difference across in-gap, gap-to-QL and within-QL configurations. On fixed images, foundation and fine-tuned profiles retain the qualitative shape of the in-gap reference more closely than the archived scratch profile. That relative success does not extend to the layer interior: the foundation and fine-tuned profiles assign much larger energies to the strongly distorted within-QL images. The seed-resolved matrix in Fig. 5 confirms that this discrepancy is not confined to a single displayed profile.

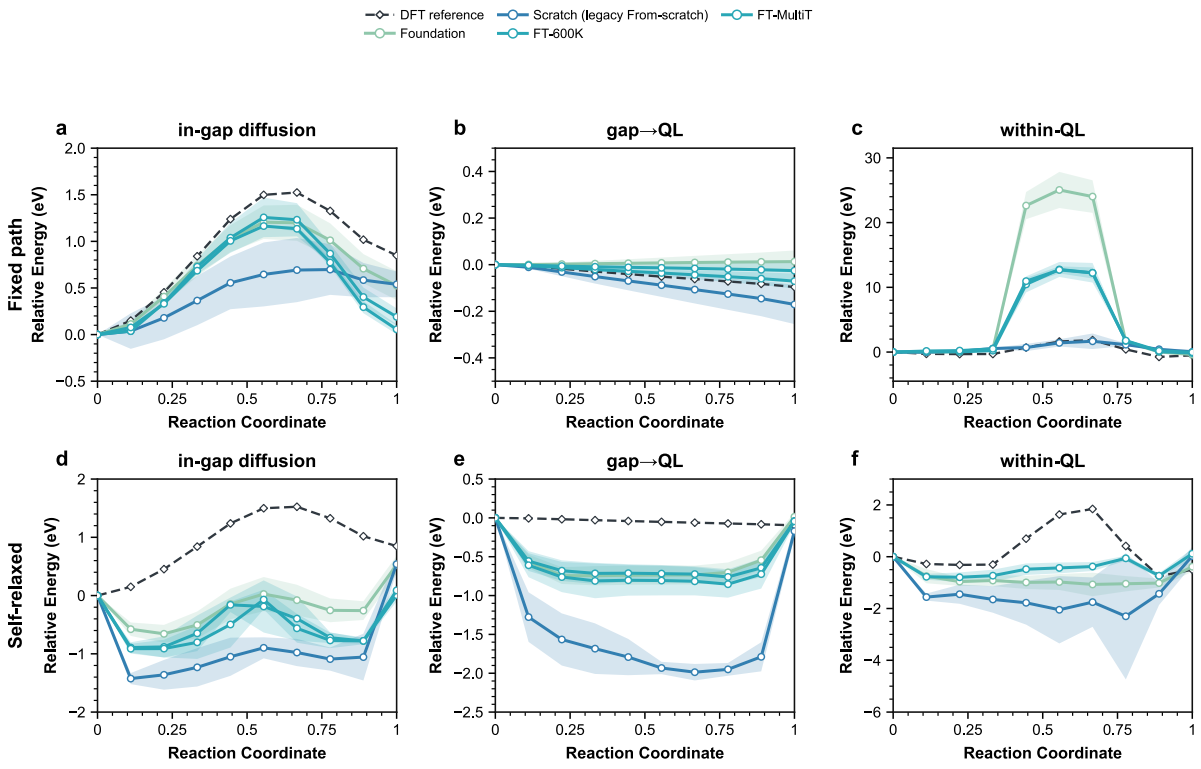


Fig. 4 Fixed geometries and model-relaxed paths reveal different discrepancies. a–c, Energy profiles evaluated on fixed reference images for in-gap, gap-to-QL and within-QL motion, respectively. **d–f,** Corresponding endpoint-initialized, MLIP-relaxed pathway diagnostics. Energies are relative to each profile's first image; the horizontal coordinate is normalized image index and does not imply that optimized paths follow the same geometric arc. Dashed dark curves are the DFT reference profiles. Coloured curves and bands summarize the archived model groups and their pointwise variation. The scratch curve is explicitly the legacy scratch condition, rather than the complete contemporary Scratch-600K/MultiT seed matrix in Fig. 5; ensemble membership is documented in Supplementary Table S3. Missing model profiles indicate unavailable or excluded accepted records. The reference peaks are 1.5243, 0 and 1.8449 eV for the three columns. Their source records retain SCF warnings, so they are image-set energy diagnostics. Self-relaxed profiles are endpoint-only MLIP NEB proxies and do not establish DFT-validated minimum-energy paths.

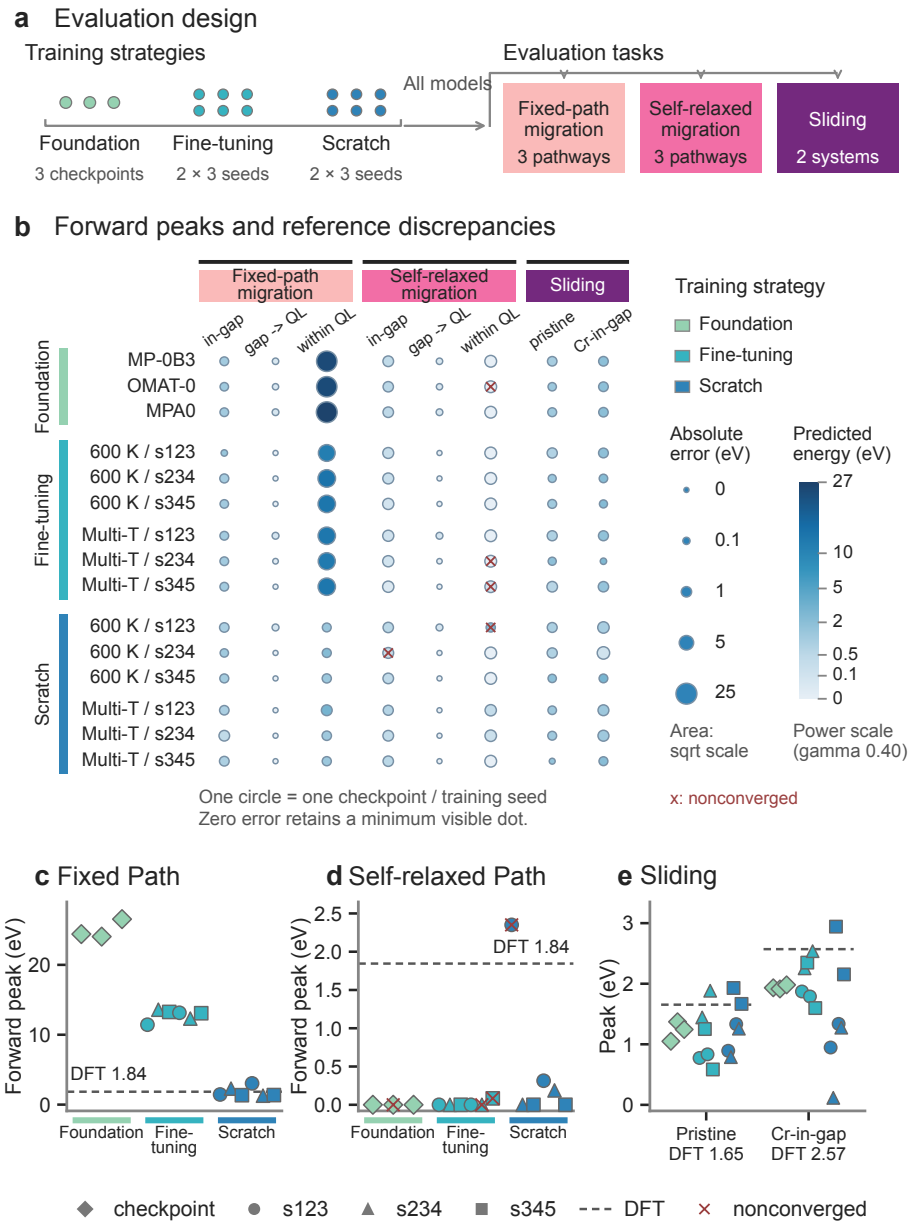


Fig. 5 Reliability depends on the task and evaluation operator. a, Evaluation design: three foundation checkpoints and three training seeds for each of two fine-tuned and two scratch conditions, evaluated over three fixed migration paths, three self-relaxed migration paths and two fixed sliding paths. **b,** The 120-record source matrix. Colour encodes the predicted forward peak relative to the first image; circle area follows a square-root mapping of the absolute discrepancy from the reference. Colour uses a power normalization with exponent 0.40. A minimum dot remains at zero discrepancy. **c,d,** Within-QL forward peaks for fixed and self-relaxed paths; the reference line is 1.8449 eV and the vertical scales differ. **e,** Fixed-path sliding peaks for pristine and Cr-in-gap systems, with reference peaks of 1.6538 and 2.5708 eV. Shapes identify training seeds or foundation checkpoints. All finite values are retained as diagnostics. Crosses identify five nonconverged records; the remaining source tags comprise 75 finite-diagnostic and 40 optimizer-converged records. A hash-matched raw-record correction changes the Scratch-600K seed-123 within-QL self peak from 0 to 2.351 eV and corrects two convergence labels (Supplementary Sec. 2.2). Neither a small dot nor a small self-relaxed peak certifies a physical barrier. Migration references retain SCF warnings, and fixed sliding endpoints and profiles require separate consistency checks. Numerical values, statuses and the correction audit accompany the figure.

On the selected within-QL image set, the reference peak relative to the first image is 1.8449 eV. In the seed-resolved comparison (Fig. 5), the three zero-shot checkpoints assign 24.04–26.54 eV and the six fine-tuned checkpoints assign 11.44–13.58 eV. Scratch-600K gives substantially smaller absolute reference discrepancies, 0.45 ± 0.05 eV, across its three seeds (Supplementary Table S4). This lower fixed-image discrepancy is reproducible across the tested Scratch-600K initializations; it should not be dismissed as a chance result, although it does not establish a reliable self-generated pathway. Those same scratch training conditions do not sustain the MD continuation.

The in-gap comparison illustrates why a statement about fine-tuning must identify its comparator. FT-600K has a mean absolute reference discrepancy of 0.27 ± 0.21 eV in the seed-resolved comparison, compared with 0.44 ± 0.16 eV for Scratch-600K; the three zero-shot values span 0.31–0.38 eV. The small number of seeds and checkpoint differences do not justify a general ordering from these overlapping ranges. The material, image set and checkpoint lineage must therefore remain attached to each result.

Letting a model relax the path changes the question. Several optimized profiles have deep intermediate minima or small forward peaks while missing the shape of the reference. Most within-QL forward peaks in Fig. 5d collapse towards zero despite the large fixed-image discrepancies. Such a reduction is not evidence that relaxation has corrected the energy landscape. It can also result from changing the sampled path, basin or endpoint relationship. A finite or optimizer-converged band still requires structural correspondence and DFT validation before it can support a kinetic interpretation.

The reference quality sets a further limit. The canonical ten-image migration references used in Figs. 4 and 5 retain electronic-convergence warnings in their source records. Their reported peaks are therefore diagnostic image-set references, not validated activation energies. A zero forward peak for the downhill gap-to-QL reference likewise tests neither the reverse event nor the complete profile. We retain these probes to expose large disagreements and operator dependence, rather than to infer migration rates.

2.4 Collective sliding tests what local adaptation leaves unconstrained

Relative layer displacement changes registry throughout the structure. It therefore probes a different response from moving a single Cr atom, even though each task occurs in the same host. Figure 5e and Supplementary Fig. S10 compare pristine and Cr-containing sliding profiles. The predicted peaks vary substantially across seeds, and some approach a reference peak while failing to reproduce the profile or the nominal endpoint return.

This distinction is particularly visible in the fixed-path curves. The DFT sliding profiles return close to their initial reference energy, whereas several MLIP curves retain an endpoint offset. A matching peak can coexist with an incorrect registry response. Such an offset must first be checked against the atom mapping, periodic displacement and calculator consistency; it is not, by itself, evidence that a model has lost long-range physics. Similarly, a coordination descriptor or a finite cutoff does not identify which interaction caused the discrepancy.

The joint result is a task-specific trade-off. A training route with a smaller discrepancy on one distorted image set can fail under its own dynamics. A checkpoint with a long-lived trajectory can misassign a migration energy. Sliding adds a profile and endpoint-consistency requirement that neither of those tests supplies. These observables should remain separate until a particular application provides a reason to combine them.

2.5 Projection geometry is a diagnostic, not a mechanism

We next ask whether the generated trajectories offer a useful view of where model conditions differ. Figure 6 visualizes 927 sampled frames across the 15 MD models using t-SNE, PHATE and UMAP [12–14]. The projected features describe recorded energies, forces and structural summaries; they are not native MACE hidden states, so their separation cannot establish distinct learned physical mechanisms. The sampled populations also differ in duration because the scratch trajectories terminate early.

There is a quantitative reason to resist a stronger reading. For the current 927-frame dataset, the unprojected output-feature silhouette is 0.063 and becomes -0.059 after feature standardization, compared with 0.117 in the displayed two-dimensional t-SNE projection. Repeating t-SNE across three perplexities and three nominal seeds gives projected silhouettes of 0.079–0.133 (Supplementary Fig. S12). Raw model energies have different reference offsets, and the archived SOAP summaries were fitted with separate PCA bases. We therefore also compute separation on the common 30-bin neighbour histogram, which gives 0.056 before standardization. None of these values supports a claim of disjoint local-geometry manifolds.

The useful role of the projections is narrower. They identify trajectory populations and motivate checks of the underlying structures, scales and failure horizons. Their connection to a physically relevant

error must be established with the process observables themselves, rather than inferred from the distance between coloured clusters.

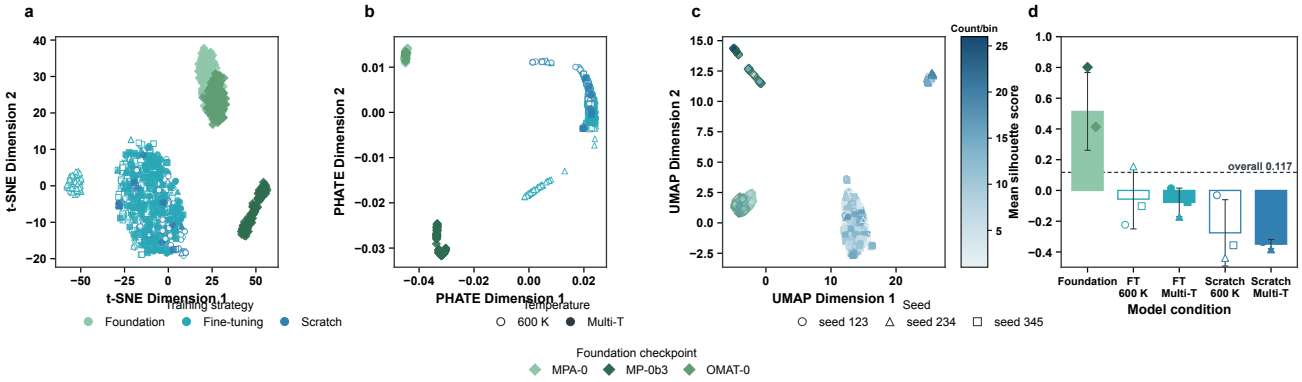


Fig. 6 Trajectory-feature projections across the 15-model MD comparison. a, t-SNE with perplexity 30 and random state 42. **b,** PHATE with random state 42. **c,** UMAP with 15 neighbours, minimum distance 0.1 and random state 42; colour intensity indicates local two-dimensional bin count on a 45-bin histogram, not energy or physical probability. **d,** Per-condition means of model-specific silhouette scores computed in the t-SNE plane, with individual checkpoint/seed symbols and a dashed overall value of 0.117. These are projection-space scores, not original-space separability or independent thermal-replica uncertainty. The input contains 927 frames, with 4–101 frames per model, and combines energies, Cartesian forces, neighbour histograms and per-trajectory SOAP/PCA summaries. Model energy offsets, feature scale, separate SOAP PCA bases and unequal failure horizons limit geometric interpretation. Original-space and projection-sensitivity checks on the same dataset are reported in Supplementary Fig. S12 and Supplementary Table S9. Constant columns are excluded from the quantitative distance analysis.

2.6 Error surrogates reveal associations and their instability

To connect structural features to a reference error, we use a separate, common held-out trajectory. Each trained model is evaluated on the same 267 configurations, and a gradient-boosted surrogate predicts its absolute energy error per atom from 390 SOAP and 17 structural features. TreeSHAP then attributes the surrogate prediction to those inputs [15]. This common-configuration comparison avoids confusing differences in sampled MD populations with differences in error sensitivity.

The current cross-validation diagnostic gives R^2 values of 0.981 ± 0.003 for FT-600K, 0.973 ± 0.003 for FT-MultiT, 0.976 ± 0.006 for Scratch-600K and 0.952 ± 0.022 for Scratch-MultiT (Supplementary Table S11). These values describe the five-fold diagnostic surrogate across the three training seeds, not independent extrapolation to a new trajectory. The final, higher-capacity TreeSHAP estimator is fitted on the full common sample; its in-sample fit does not validate generalization.

Figure 7 retains seed identity because the feature ranking varies substantially. Te–Te and Cr-containing descriptors appear among the leading fine-tuned features, while scratch conditions emphasize a different combination of host and dopant environments. Several between-seed standard deviations are comparable to or larger than the corresponding means. This variability argues against identifying a single dominant Cr–Cr channel or assigning the observed pathway behaviour to one internal feature.

A supplementary coordinate-perturbation study tests the local relaxation response of native MACE inputs across six dopant–host combinations. It supplies a physical-input robustness check, but does not isolate a SOAP channel or establish causal attribution for migration errors.

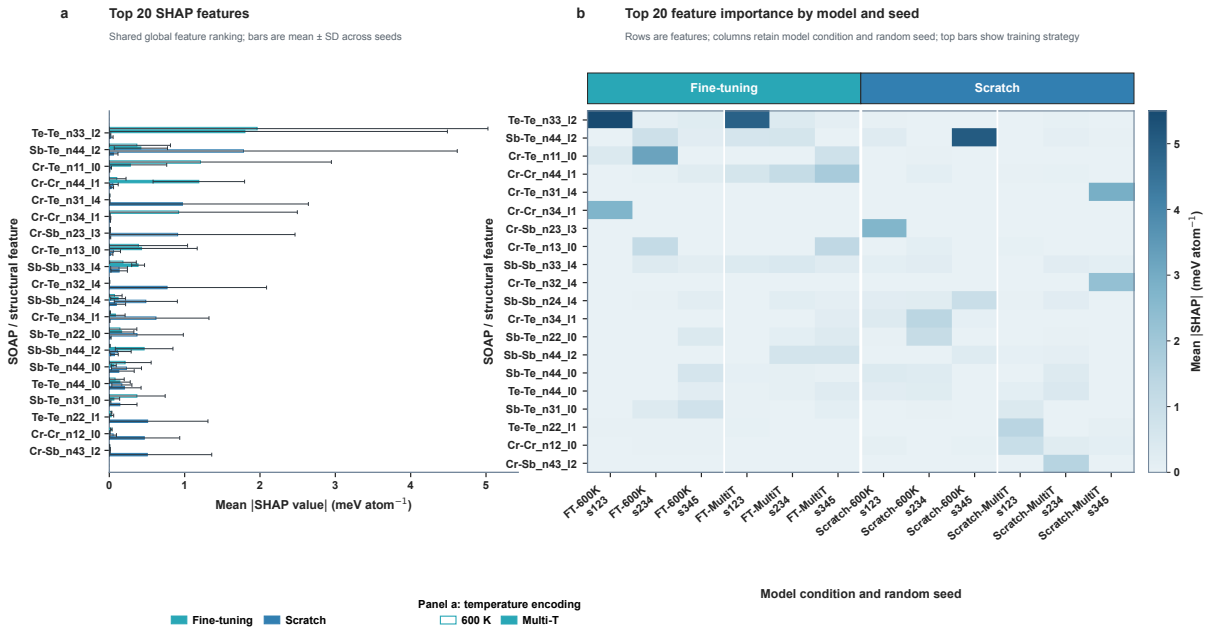


Fig. 7 Structural predictors of energy error depend on model condition and training seed. TreeSHAP explains gradient-boosted surrogates of the absolute energy error per atom on one common 267-frame test trajectory. **a,** Mean absolute SHAP values for the 20 features selected by the shared global ranking, with mean and sample standard deviation over training seeds 123, 234 and 345 within each of four conditions. Strategy colour distinguishes fine-tuning and scratch training; open/filled bars distinguish 600 K and MultiT. **b,** The same 20 feature rows aligned across all 12 model–seed columns on a shared meV atom^{-1} scale. Species pairs and n_1n_2, l identify SOAP power-spectrum channels; they are descriptor indices, not isolated bonds or MACE neurons. Surrogates use 407 computed features and frame-aligned errors. Fidelity of the separate five-fold diagnostic estimator is given in Supplementary Table S11; it does not make the full-data TreeSHAP fit an out-of-sample causal explanation.

3 Discussion

When does a foundation model help? In this study, the answer changes with the observable. The tested zero-shot checkpoints sustain the longest MD continuations, while target adaptation reduces held-out force error and improves selected fixed-image energy comparisons. Neither benefit supplies the other. Within-QL distortions expose large energy discrepancies in foundation and adapted models, while scratch models with smaller discrepancies fail rapidly in MD. The central lesson is that a transferable prior and a trustworthy simulation are different objects: the former becomes useful only where the latter has been established for the intended process.

These results suggest a practical order for making that decision. A candidate model should first be checked against the target electronic-structure reference on common configurations, including the coordination changes that matter to the process. It should then be allowed to generate short paths or trajectories, with convergence, geometry and stopping criteria inspected directly. Collective displacements and endpoint-return checks can test responses that local thermal sampling leaves unconstrained. Passing one stage provides evidence for that stage; it does not certify the remaining observables.

Distribution shift is a useful way to organize this reasoning, but its meaning must be specific. The adaptation dataset can miss a process-relevant geometry even within the same chemistry and temperature range. Model-driven trajectories can create a further shift by visiting configurations absent from the reference set. Differences between the pretraining and target electronic-structure protocols can add calibration errors. We do not have a complete, matched audit of the foundation pretraining distributions, so the present results do not identify a numerical pretraining-distance threshold or prove that distribution shift is the sole cause of failure.

This distinction also gives active learning a more precise objective. A promising next step is to acquire DFT labels where process-relevant displacements or short model rollouts reveal large errors, unstable forces or unresolved reference profiles. Updating a model should then be judged by improvement on held-out process tests while rechecking the behaviours that previously worked. Such a loop could guide the choice between adaptation and a specialist, and constrain how adaptation is performed. Feature attributions or model disagreement may help nominate configurations, but their acquisition value requires a prospective comparison under a matched DFT budget.

The conclusions concern the tested MACE routes in Cr-doped Sb_2Te_3; additional dopants, hosts and controlled architecture-matched comparisons are needed before treating the workflow as a general model-selection standard. The diagnostic references with convergence warnings further preclude quantitative kinetic claims. Within those limits, the tests answer a useful experimental question: a foundation model helps when it supplies accuracy or robustness that survives the specific process being simulated, and fine-tuning helps only when that benefit survives the same scrutiny.

4 Methods

4.1 Reference data and model lineages

The reference archive contains PBE-based Quantum ESPRESSO calculations and AIMD data for Cr-doped Sb_2Te_3 [16, 17]. The target training pools include 600 K and multi-temperature configurations; path-specific computational settings and convergence states are distinguished in Supplementary Sec. 1. We do not combine warning-bearing migration references with converged paths to obtain a pooled kinetic score.

The contemporary pointwise and MD comparison contains MPA-0, MP-0b3 and OMAT-0 foundation representatives and three seeds for each trained condition. Fine-tuned grouped-A2 models use an OMAT-FT starting checkpoint; scratch models are the registered $L_{\max} = 1$ implementations. The source release also contains legacy pathway checkpoints, which are identified separately in Figs. 4 and 5. Family names describe training routes and do not imply identical architectures, initialization checkpoints or hashes across all archived campaigns.

Training, validation and test partitions are assigned by complete source trajectory for the grouped comparisons, so adjacent frames from one source do not cross partitions. The 600 K grouped split contains 1,786/311/267 configurations. The MultiT pointwise evaluation contains 2,659 training and 767 test configurations. An additional overlap audit of 25 legacy migration images finds no exact or threshold-defined SOAP near-duplicate in the declared training files; its scope is that audited image set, not every subsequently assembled path. The matched 600 K scratch and fine-tuned routes use the same 1,786 target training configurations; the separate 89-configuration Scratch-5% control is a data-volume sensitivity check, rather than a data-matched estimate of the effect of pretraining. Architecture and optimization remain part of each route.

The fine-tuning provenance audit finds no NEB-derived or unclassified all-zero-force training frame among 9,163 audited records; the nine exact-zero-force records are labelled one-atom energy-reference structures. Missing-force NEB exports are excluded from force-supervised training. The supporting audit retains the source-file identities and does not imply that every later campaign has the same training provenance.

4.2 Process observables and uncertainty

For a reference image set $\Gamma_{\mathrm{ref}} = \{X_i\}$, the fixed-path forward peak is

$$B_M^{\mathrm{fixed}}(\Gamma_{\mathrm{ref}}) = \max_i \{E_M(X_i) - E_M(X_0)\}. \tag{1}$$

The diagnostic discrepancy is $|B_M^{\mathrm{fixed}} - B_{\mathrm{ref}}|$. It is an activation-energy error only when the electronic reference, pathway optimization and endpoint definition support that interpretation. A self-relaxed value uses $\Gamma_M = \mathrm{NEB}_M(X_0, X_1)$ instead. Comparing those values therefore changes both the energy function and the sampled configurations. Full profiles and status records are retained alongside peak values.

The MD comparison uses a common 200 ps starting state, 1 fs timestep, 200-step trajectory interval, 600 K thermostat target and MD seed 1100. Sample standard deviations across training seeds describe checkpoint variation; foundation variation is across distinct checkpoints. Neither estimates uncertainty over independent starting-state or velocity replicas. Observed stop times, rather than extrapolated continuations, determine the usable windows.

4.3 Diagnostic analysis and reproducibility

The trajectory analysis uses the sampled output-feature matrix. The quantitative distance analysis excludes constant columns and contains no force-sensitivity channel. It calculates original-space distances before and after standardization, and repeats t-SNE at perplexities 5, 30 and 50 with nominal seeds 0, 42 and 123. A common neighbour-histogram analysis avoids treating independently fitted SOAP PCA coordinates as one shared physical basis. The SHAP analysis uses a separate common held-out configuration set, not

the sampled MD feature matrix. Surrogate definitions, split limitations and complete per-seed fidelity metrics are provided in Supplementary Sec. 6.

Data availability

The figure-associated numerical tables, model and dataset identities, convergence-status records and current diagnostic outputs accompany the submission source package. The project repository is https://github.com/yicao-elina/MigrationBench. Figure-specific source identities are provided with the Supplementary Data; a family label alone is not a checkpoint identifier.

Code availability

The accompanying source package includes the executed scripts and frame-aligned inputs for the current SHAP analysis, the current feature audit, and the scripts and compact source tables used to assemble the figure evidence.

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Author contributions

Y.C.: conceptualization, methodology, data curation, DFT and MD simulations, analysis, visualization and writing. P.M.: literature review, conceptualization and writing. P.C.: conceptualization, project administration, supervision, funding acquisition and writing.

Competing interests

The authors declare no competing interests.

References

[1] Unke, O.T., Chmiela, S., Sauceda, H.E., Gastegger, M., Poltavsky, I., Schütt, K.T., Tkatchenko, A., Müller, K.-R.: Machine Learning Force Fields. Chemical Reviews **121**(16), 10142–10186 (2021) https://doi.org/10.1021/acs.chemrev.0c01111

[2] Batzner, S., Musaelian, A., Sun, L., Geiger, M., Mailoa, J.P., Kornbluth, M., Molinari, N., Smidt, T.E., Kozinsky, B.: E(3)-equivariant graph neural networks for data-efficient and accurate interatomic potentials. Nature Communications **13**(1), 2453 (2022) https://doi.org/10.1038/s41467-022-29939-5

[3] Batatia, I., Kovacs, D.P., Simm, G., Ortner, C., Csányi, G.: Mace: Higher order equivariant message passing neural networks for fast and accurate force fields. Advances in neural information processing systems **35**, 11423–11436 (2022)

[4] Batatia, I., Benner, P., Chiang, Y., *et al.*: A foundation model for atomistic materials chemistry. The Journal of Chemical Physics **163**(18), 184110 (2025) https://doi.org/10.1063/5.0297006

[5] Deng, B., Zhong, P., Jun, K., Riebesell, J., Han, K., Bartel, C.J., Ceder, G.: CHGNet as a pretrained universal neural network potential for charge-informed atomistic modelling. Nature Machine Intelligence **5**(9), 1031–1041 (2023) https://doi.org/10.1038/s42256-023-00716-3

[6] Alghamdi, N., Angelis, P., Asinari, P., Chiavazzo, E.: Comparing fine-tuning strategies of mace machine learning force field for modeling li-ion diffusion in lif for batteries. arXiv preprint arXiv:2510.05020 (2025)

[7] Bheemaguli, A.K., Xiao, P., Sai Gautam, G.: Evaluation of foundational machine learned interatomic potentials for migration barrier predictions. Digital Discovery **5**, 1809–1819 (2026) https://doi.org/10.1039/D5DD00534E

[8] Deng, B., Choi, Y., Zhong, P., Riebesell, J., Anand, S., Li, Z., Jun, K., Persson, K.A., Ceder, G.: Systematic softening in universal machine learning interatomic potentials. npj Computational Materials **11**(1), 9 (2025) https://doi.org/10.1038/s41524-024-01500-6

[9] Zhang, H., Liu, C.-X., Qi, X.-L., Dai, X., Fang, Z., Zhang, S.-C.: Topological insulators in Bi2Se3, Bi2Te3 and Sb2Te3 with a single Dirac cone on the surface. Nature Physics **5**(6), 438–442 (2009) https://doi.org/10.1038/nphys1270

[10] Cortie, D., Zhao, W., Yue, Z., Li, Z., Bake, A., Marenych, O., Pastuovic, Z., Nancarrow, M., Zhang, Z., Qi, D.-C., Evans, P., Mitchell, D.R.G., Wang, X.: Creating thin magnetic layers at the surface of Sb2Te3 topological insulators using a low-energy chromium ion beam. Applied Physics Letters **116**(19), 192410 (2020) https://doi.org/10.1063/5.0006447

[11] Cao, Y., Clancy, P.: Migration as a probe: A generalizable benchmark framework for specialist vs. generalist machine-learned force fields. arXiv preprint arXiv:2509.00090 (2025). AI for Accelerated Materials Discovery (AI4Mat) Workshop, NeurIPS 2025; non-archival workshop report

[12] Maaten, L.v.d., Hinton, G.: Visualizing data using t-sne. Journal of machine learning research **9**(Nov), 2579–2605 (2008)

[13] Moon, K.R., Van Dijk, D., Wang, Z., Gigante, S., Burkhardt, D.B., Chen, W.S., Yim, K., Elzen, A.v.d., Hirn, M.J., Coifman, R.R., *et al.*: Visualizing structure and transitions in high-dimensional biological data. Nature biotechnology **37**(12), 1482–1492 (2019)

[14] McInnes, L., Healy, J., Melville, J.: Umap: Uniform manifold approximation and projection for dimension reduction. arXiv preprint arXiv:1802.03426 (2018)

[15] Lundberg, S., Lee, S.-I.: A Unified Approach to Interpreting Model Predictions. arXiv (2017). https://doi.org/10.48550/arXiv.1705.07874

[16] Giannozzi, P., Baroni, S., Bonini, N., Calandra, M., Car, R., Cavazzoni, C., Ceresoli, D., Chiarotti, G.L., Cococcioni, M., Dabo, I., *et al.*: Quantum espresso: a modular and open-source software project for quantumsimulations of materials. Journal of physics: Condensed matter **21**(39), 395502 (2009)

[17] Perdew, J.P., Burke, K., Ernzerhof, M.: Generalized gradient approximation made simple. Physical review letters **77**(18), 3865 (1996)